An effect of ICT on economic growth of developed and developing countries

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Abstract

This paper examines the effect of information communication and technology (ICT) on the impact of economy of a country, especially prioritizing the context of Nepal. This study is synthesized from prior studies and goes over the effect of Information communication and technology in business industries, the effect on productivity and ultimately examining the economy in general, comparing the study from cross-country analysis, as well as comparisons between the income groups. The comparison of multiple papers shows that although there may not exist a direct linear relation, it can be inferred from the positive effect on the labour productivity growth and improvements of sales in businesses that theres is a net positive effect of Information communication and technology in the economy of the country but the magnitude og growth may differ from high and low income countries.

1 Introduction

Mobile phones, the Internet, and other means of communication have been a major part of the world, connecting us all globally, and we have become more reliant on information, technology, and communication than ever before. In the last decade, Nepal has also improved tremendously in terms of information, communication, and technology. Since the establishment of the National Technology Center in 2001, the government of Nepal has been promoting ICT growth. With the use of computers in various sectors, the amount of financial spending on computing technologies has also rapidly increased (Haimowitz, 1998), which leads us to question whether or not communication and information sharing in these sectors could have a significant impact on the economy of any country. The rise of computers in businesses and other private sectors has led some analysts to suggest that the economy is entering a new era (Haimowitz, 1998) . Since the world around us is surrounded by computers and other forms of communication devices, these devices and forms of communication are seen as the key drivers of productivity growth, and productivity growth is the foundation for improving the living standards of people. Looking at the World Bank's data on the number of people using the internet, we can see that compared to the year 2000, when the percentage of people using the internet was just about 7% of the world population, the number has increased by more than 56% to currently be more than 63% of the world population using the internet. Similarly, the rise in mobile cellular subscriptions per 100 inhabitants has gone up from 12 to 108 in 2022 (World Bank, 2024a, 2024b).tion communication and technology in business industries, the effect on productivity and ultimately examining the economy in general, comparing the study from cross-country analysis, as well as comparisons between the income groups.

2 Introduction

Comparing the few countries ranked as the most developed in ICT, according to the world digital competitiveness index of 2023, the USA, Netherlands, and Singapore are at the top. The report shows that in 2023, the USA increased the number of 4G and 5G connections in the mobile market from 86% to 96%, while wireless penetration per 100 people increased from 159 to 163. Similarly, the Netherlands and Singapore have all improved in various factors by varying percentages. (Center, 2023)

So ICT makes available a much more effective and efficient transfer of information, new communication methods, and improved productivity and effectiveness in the company and the economy as a whole (Bilan et al., 2019). There have been tremendous improvements in the field of information technology and communication in recent times, and with that improvement, its impact on the country's economy is also seen. As stated by (Kurniawati, 2021),investments in telecommunication technologies have significantly promoted growth and made ICT one of the more significant drivers of economic growth. Past transformative technologies like electricity took decades for their full economic impact to be seen, so the question still remains whether the "New Economy" for ICT is yet to come (Haimowitz, 1998). Therefore, the study of ICT and its effect on the country's economy becomes extremely necessary in order to understand the current economic trajectory of the country and to look into what can be done to further improve the direction a country might be headed.

ICT has been a contributing factor, like improving drinking water and public transportation and helping enhance labor productivity, which could facilitate economic activity (Maneejuk & Yamaka, 2020). Information, communication, and technology can produce potentially significant gains in developing countries (Blinder & Quant, 1997), which makes this study even more important for developing countries like Nepal. But as stated in a paper by Raéf Bahrini and Allah Qaffg , there have been many studies that have looked into the impacts of ICT on economic growth, and although many papers show that there is a significant impact, many empirical studies have produced many mixed and inconclusive results (Bahrini & Qaffas, 2019). Other research has found the impact of ICT to be a multidisciplinary field that includes information systems, human-computer interactions, communication studies, and developmental studies (Avgerou, 2010).

Much of the literature has a strong focus on some key indicators like Internet penetration, mobile phones, broadband connections, and telephone line establishment, which would not take into account the broad discipline that ICT relates to. With the implementation towards the goals of Nepal digital framework made in recent years Nepal has also seen a lot of improvement In ICT services National Information Technology Center was established in 2001 with the objective of facilitating ICT enabled government services. Many of the researches and articles that look into development of ICT and ICT related topics of the developed countries and developing countries does not include the context of Nepal and very few literature is found for ICT development in Nepal. This study aims to look into many empirical as well as literature review and make a comparative study in order to generate a much more suitable conclusion and use it to see what improvements we can make in the current policy that is in effect in Nepal to develop the country's ICT foundation. It is important to note that ICT adoption alone doesn't guarantee economic growth especially for developing countries like Nepal must strategically employ ICT while addressing the foundational developmental challenges.

3 Methodologies and Literature Search

In this study, we systematically review all comparable research into ICT and its relation to economic growth and perform analysis to see how each research is similar or dissimilar to each other. We look into the various types of indicators used in both empirical and non-empirical studies and generate a list of ideas and key points that could lead us to the proper conclusion. We also look into policies, policy literature, and publications in order to recommend policies that take all of our research and studies into consideration. Since the general view in economics is that ICT has a positive effect on the economy, our focus lies on finding the overall contribution of ICT and ways countries like Nepal can adapt to the changing ICT economy. The starting point of our review was to search for various papers related to the topic. We started by looking for cross-country studies and some country-wise studies. After sourcing the papers, we begin by categorizing the results of similar papers and have found that newer papers tend to find a more positive correlation compared to older papers. We examine each paper in depth. We independently searched for keywords in Google Scholar and searched for the works listed in that paper.

3.1 ICT Policy Analysis Methodologies

The policy analysis follows a six step process from initially listing the policies in context and checking for its effectiveness. Has the goal of policy been achieved and if it has how effectively has it achieved the desired goal.

4 Literature Review

Over the past decade, there have been extensive studies related to information, technology, and its impact on the economy. A significant body of literature has been published analyzing how investments in ICT, including computers, internet infrastructure, etc., impact the productivity and Efficiency of the economy (Bilan et al., 2019; Haimowitz, 1998).

4.1 ICT in Business Industry

The case study of Ukraine (Bilan et al., 2019), where the authors analyzed two hypotheses, one where a high level of ICT usage in business leads to the formation of a competitive advantage and accordingly the growth of their financial and economic results, and another where a high level of ICT development leads to the creation of added value and accordingly GDP per capita growth. By testing the use of information, communication, and technology in various companies, the authors have found a positive correlation between sales volume and that all the indicators that use ICT have a significant direct impact. To analyze and predict the impact of ICT on economic development, the authors establish a connection between the indicators of ICT at different companies with different economic activities. And with that, the authors of the paper predict the value of sales growth by up to 10.41% and expect a GDP growth rate of 4.17%. The author states that the Information and Communication Technology Development Index and the ICT component of the global competitiveness index confirm the existence of a positive correlation between ICT and GDP per capita. Looking at the case study of Ukraine, we can safely conclude that ICT does have some level of positive impact on sales volume when businesses give priority to ICT technologies. This study also aligns with the paper by Isaac Appiah-Otoo and Na Song, where they discuss how ICT promotes economic growth by improving business and other factors of quality of life and competitiveness.

4.2 ICT in Productivity Growth

The paper by Federico Biagi (2013) looks into ICT and its effect on productivity and growth (Biagi, 2013). The paper gives importance to four major types of ICTs: computers, software, telecommunication equipment, and semiconductors. The paper states that ICT is said to directly and indirectly affect growth and productivity, especially with the use of the internet, computers, and other technological improvements. There are several theories proposed in the paper. Since ICT plays a huge role in product, process, and organizational innovation, they can be called GPT, or general purpose technologies, and such technologies can be used on production processes, which allows for continuous improvements, making ICT an important factor in growth and productivity. The paper also states that ICT's productivity-enhancing effect may not be visible immediately or may not be direct, but rather is mediated by the development of other technologies and thus is able to generate an increase in productivity and the economy. The paper finds that while there are exceptions, many other papers reviewed interpret ICT as a GPT. While the overall literature suggests that ICT has without a doubt highly contributed to productivity and growth, the author states that there is still no conclusive evidence for the measurement and interpretation of such effects. Reviewing the impacts of ICT and productivity in the late 1990s, when the US experienced a period of rapid economic growth, the impacts of ICT were seen to be clear, which is said to have accounted for two-thirds of the one-percent step-up points in productivity growth. The paper finds that the contribution of ICT's capital to labor productivity increased by a significant margin and reached 48% in 1994–1998. But despite this, labor productivity slowed in the UK after 1994, unlike the accelerated economy of the United States. Although the exact reasons for such divergence in total factor productivity are not clear, overestimation of IC returns or high adjustment costs might be some potential factors. After adjusting the prices for the UK's GDP for US ICT prices, software investments show that the GDP growth is higher, especially since 1994, and ICT accounted for more than 20% of growth in GDP in the 1990s. The ICT income share in the UK is only about two-thirds of that in the US, suggesting that the ICT contribution may increase further. An empirical analysis (Cardona et al., 2013) looks into direct outputs like GDP at the country level and productivity and sales at the firm level. Just as stated above, the most common way to measure productivity is labor productivity. The paper compares some major ways of looking into productivity, like growth accounting and productivity estimation methods. Growth accounting provides a well-established tool for how much of a country's or industry's growth in outputs can be explained by growth in inputs, especially by growth in various types of capital inputs. The paper states that the main critique of growth accounting is that it does not explicitly account for the underlying causes of growth. The paper also looks into other forms of analysis, like productivity estimation and testing for Granger causality. The growth accounting studies show that there is a stronger effect of ICT on productivity in the US compared to European countries. These growth accounts suggest a positive spillover from ICT, which supports the idea of ICT as a GPT. On the contrary, the econometric firm-level studies argue that there is no significant difference in the magnitude of ICT effects between the US and other countries. According to the authors of the paper, ICT meets many criteria for the GPT, which enable innovation; however, the evidence regarding the ICT spillover affecting the total factor productivity is mixed. Some industry studies have shown higher productivity growth in ICT-intensive sectors, while others show no spillover to total factor productivity. Although the paper highlights the significant positive effect of ICT that has been found in productivity growth, which seems to be rising over time, it suggests that further studies are needed to conclusively demonstrate its spillover effect. Overall, we can say that although ICT appears to improve productivity growth, especially with the growth accounting approach, we cannot conclusively state that ICT improves productivity, and further research is to be done.

4.3 ICT in Economic Growth

Technological progress has improved tremendously over time, and with that, its impact on the country's economy. Currently, ICT indicators are considered essential to evaluating the competitiveness of a country and are said to have a positive correlation with the GDP of a country (Bilan et al., 2019). To understand this much more clearly, we need to perform cross-country studies along with some country-level studies. The paper by Isaac Appiah-Otoo and Na Song looks into the comparative study between the ICT of developed and developing countries (Appiah-Otoo & Song, 2021). The paper has focused on three major indicators to evaluate economic growth: mobile phone, internet availability, and fixed broadband. The paper states that ICT promotes economic growth through improvements in quality of life, improved competitiveness in doing business, diversification of the economy, and retention of business. Along with diffusion in technology, learning, and innovation, improved decision-making quality and cost reduction, and decisions that help improve both the demand and supply curve, Transaction costs in financial services are drastically reduced by ICT. The paper also states that mobile phones have supported economic growth in many Indian states, and countries like Singapore and Australia are found to have increased economies with the increase in ICT infrastructure. Many countries, including Finland, the USA, Sweden, and Korea, have also seen a positive increase in their economies with the increase in ICT adoption. Research that examines the impact of ICT and innovation on carbon emissions and the economic growth of G-20 countries has found that ICT does spur economic growth. Many cross-country studies, like (**<empty citation>**) research that looked into the ICT infrastructure in EU countries, have shown increased economic growth with the development of the ICT industry. The author finds that ICT does, in fact, have a significant positive impact on the economy of the country. Mobile phones, internet, and fixed broadband have an impact of 0.26%, 0.75%, and 0.32%, respectively, on the country's economic growth. The middle-income countries have seen an increase of 0.44%, 0.35%, 0.19%, and 0.42% in mobile phones, internet, and fixed broadband. And the low-income countries have growth of 0.09%, 0.15%, 0.09%, and 0.16%, all at the 1% significance level. The significant positive results coincide with another finding, (Myovella et al., 2019), looking into 33 OECD and 41 SSA countries that also find ICT to contribute to economic growth. The results of the research show that high-income countries gain more than middle- and low-income countries in terms of fixed internet adoption and fixed broadband. The paper suggests that the reason that high-income countries could have higher gains could relate to high-level investments, which is again backed by Myovella (2020). The paper also finds that the growth effect of mobile was highest in MIC. This implies that MIC gains more than HIC and LIC from mobile phone adoption. Another paper by Paravee Maneejuk and Woraphon Yamaka that analyzes the impacts of telecommunication technologies and innovation on economic growth at two different levels (country level and cross-country level) has found that there exists a kink effect in relation to telecommunication technologies and the economic growth in both developing and developed countries, and depending on the level and kink points, some related variables differ in magnitude, and telecommunication technologies have a relatively greater impact on the economic growth of developing countries compared to developed countries (Maneejuk & Yamaka, 2020). Most of the literature reviewed in this paper too has found positive correlation between ICT and the economy, however the paper states that many empirical researches have inconclusive results. Some literature contradict each other as stated by Paravee Maneejuk and Woraphon Yamaka, some literature show positive impact of ICT for the developed countries but nothing significant for the developing countries. While other literature shows positive correlation for developed, developing and emerging countries (Appiah-Otoo & Song, 2021). The authors suggests that the impact of ICT, on the economy may not be linear as thought previously. The author states "The different telecommunications technology and innovation indicators introduced different impacts in terms of magnitude and sign. Comparing the test results of the developed and developing countries, we find that the impacts of telecommunications technology and innovation on economic growth are slightly different in magnitude; however, the coefficients are most significant for the developing countries. This implies that telecommunications technology and innovation contribute a larger impact on developing countries compared to developed countries". Although developed countries are the first to develop and design new technologies, these same technologies have bigger impact on the developing countries.

5 Discussion

As we see how important ICT has become to the overall development and economic growth of a country, it becomes increasingly important to find out how your country is performing relative to the rest of the world. And Nepal is struggling. Although, with the establishment of the Telecommunications Act in 1997, which was again revised in 2000, Nepal began its ICT development, it still has a long way to go. According to the paper by Sharma, Ashish, and Kim, Yun Seon, the ICT infrastructure is far from satisfactory (Sharma & Kim, 2016). The telecommunications industry seems to have much slower growth rates compared to developing countries; however, the data is based on the number of SIMs sold and active, and many areas in Nepal still do not have reliable communication and 3G networks. Although the Internet and communication services in Nepal are cheaper compared to other developed and developing countries, many of these services are limited to urban areas. The paper looks into the background of Nepal's ICT development and compares it with that of neighboring countries. And it finds that while many other countries are advancing innovation and improvements in ICT services, Nepal is struggling to adopt the basic communication technologies (Sharma & Kim, 2016). In terms of the Network Readiness Index 2023, Nepal ranks 114 out of 134 economies, which suggests the current need for ICT development in Nepal (Lanvin et al., 2023). In 2019, the Ministry of Communication and Technology created a framework called the Digital Nepal Framework, which guides Nepal's digital transformation journey. However, as an article in the Himalayan Times puts it, some of the public services use technology partially, whereas most do not have access to digital platforms ("Digital transformation key to enriching people's quality of life", 2024). To improve on the digital aspect, the Nepali government created the Citizen's App, "the Nagarik App," as it is called, but the effectiveness of such an application is questionable. An article in Himalayan Times calls it a simple record keeper ("Digital transformation key to enriching people's quality of life", 2024).

The application was designed to provide many services, such as a passport, driving license, and other documents. Although the government makes sure that these documents are genuine, the government bodies themselves do not accept the documents, an article in the Rising Nepal Daily writes (Paudel, 2024). These articles show just how lacking the ICT development is. The Nepali government had also started many e-government initiatives guided by the e-government master plan created by the Korean Information Promotion Agency in 2006, which was supposed to be completed by 2011, but due to political instability and other hurdles, many components of it have yet to be implemented (Sharma & Kim, 2016), which further solidifies the need for better execution and implementation strategies. The study presents the current picture of Nepal and shows the lack of proper implementation. As per the study, Nepal needs to implement a proper execution and implementation plan, a stronger budget, and allocate some research into simulating ICT in various rural areas. Rather than just looking for more improvement in technologies, it becomes ever more necessary to increase internet and mobile phone penetration. Improving the Internet and mobile phone penetration in areas where they are lacking gives people in such areas access to services and facilities, which can improve the overall readiness for e-education. Many schools and colleges in rural areas don't have access to basic computers, which needs to be prioritized. ICT is ever-changing, and improvements and policies regarding foreign investments should also be given priority. The study also shows that the percent of budget allocation compared to ICT-rich countries is very low and ambiguous. The network readiness index ranks e-governance at 110, which shows the inability to effectively implement a strong framework. ICT development isn't solely the work of the government, and private companies should be encouraged more, as shown by Yuriy Bilan and co. in their paper, ICT and Economic Growth: Links of Possible Engagement (Bilan et al., 2019). It is important for private sectors to implement ICT in some way or form, and the government should develop policies and frameworks that make it easier for such private companies to adopt ICT.

6 Conclusion

We can thus confidently conclude that ICT has a significant effect on the economy of the country, whether that be kink effect in the relation as suggested by Paravee Maneejuk, Woraphon Yamaka, or improvement in labor productivity that causes the improvement in ICT. We have realized just how important information, communication, and technology have become in the last decade. With ICT adoption, we can clearly see growth in productivity. ICT also shows positive relations and improvements in private companies, which shows that sales growth increased by up to 10.41% after ICT adoption. Many comparative studies show that ICT does spur economic growth. Papers looking into 33 OECD countries and 41 SSA countries have also found some contribution of ICT in economic growth. Some papers comparing high-, middle-, and low-income countries show that ICT has effects that are significant in their own manner, while low-income countries tend to get more out of ICT adoption. With ICT bringing so much value to developing countries like Nepal, it is incredibly important to build policies and strategies to properly implement ICT. Improving Internet penetration and creating policies that support foreign investments in ICT will become very important in developing ICT literacy at the school level. The government should be willing to advance, so the adoption of government services online and making sure that government bodies have strong and latest connections to ICT and security become very important. Although some recommendations have been made according to the study, further research has to be done to conclusively set a direction for private institutions and plans for security policies in ICT regarding artificial intelligence and other forms of regenerative AI.

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